$$\sum_{\{j_i\}} A^{[j_1]}{}_{\mu_1} A^{[j_2]\mu_1}{}_{\mu_2} A^{[j_3]\mu_2}{}_{\mu_3} \cdots A^{[j_n]\mu_{n-1}}{}_{\mu_n} |j_1 \cdots j_n\rangle$$

$$\sum_{\alpha} \Lambda_\alpha |\alpha_L\rangle |\alpha_R\rangle$$

$$\sum_{\{j_i\}} \psi(j_1, \dots j_n) |j_1, \dots, j_n\rangle$$

will however be diagonal.

$$\sum_{\alpha_1} A^{(\mu_1)} |\alpha_1\rangle |\alpha_{2\dots n}\rangle$$

dimensionality grows; i.e. number of bond states ~ left feet

$$= \sum_{\alpha_1} \sum_{\alpha_2} A_{\alpha_1} A^{\alpha_1}{}_{\alpha_2} |\alpha_1\rangle |\alpha_2\rangle |\alpha_{3\dots n}\rangle$$

α_2 each α_2 state actually depends on α_1 state. so Λ size grows. to run through.

$\alpha_{1,2}$ has

→ continue like this

→ physical states should have limited entanglement.

Λ_α only depends on site.

$$\sum_{\{\alpha_i\}} A^{(\alpha_1)}{}_{\mu_1} A^{(\alpha_2)\mu_1}{}_{\mu_2} \cdots A^{(\alpha_n)\mu_{n-1}}{}_{\mu_n} |\alpha_1 \dots \alpha_n\rangle$$

$\langle j_i | \alpha_i \rangle$

→ introduce also change of basis matrix:

$$\sum_{\{\alpha_i\}} A^{(j_1)}{}_{\nu_1} U^{(j_1)\nu_1}{}_{\mu_2} A^{(j_2)\mu_2}{}_{\nu_2} U^{\nu_2}{}_{\mu_3} \cdots |j_i\rangle$$

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